Amelie Lam

Scientific Machine Learning | Differentiable Simulation | Fluid Dynamics | HPC

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Education

Université Côte d'Azur — *Oct 2024 – Sep 2027 (Expected)*
Doctorat en Sciences de la Planète et de l'Univers

University of Cambridge — *Oct 2020 – Jun 2024*
M.Sci. & B.A. in Natural Sciences Tripos

Research Experience

PhD Research — *Nice, France*
Observatoire de la Côte d'Azur, CNRS, Laboratoire Lagrange — *Oct 2024 – Present*

- Developing a hybrid neural network–solver framework for turbulent flow simulations, combining CNN-based subgrid corrections with differentiable numerical solvers
- Designing solver-in-the-loop training methods with multistep temporal inputs for long-term prediction of unresolved turbulent dynamics
- Developing Lagrangian particle simulations and extending the framework to rotating turbulence, with applications to planet-forming disks

Master's Research Project — *Cambridge, UK*
Institute of Astronomy, University of Cambridge — *Oct 2023 – May 2024*

- Applied deep learning methods to emulate luminosity variations in planet-forming disks and investigate dust-clump formation

Research Intern — *Cambridge, UK*
Department of Geography, University of Cambridge — *Jul 2023 – Sep 2023*

- Upgraded the cloud microphysics module of the Active Tracer High Resolution Atmospheric Model (ATHAM) from a one-moment to a two-moment scheme, predicting both hydrometeor mass mixing ratios and number concentrations
- Implemented microphysical processes including nucleation, coalescence and thermodynamic phase changes

Publications

Solver-In-The-Loop Training with Multistep Inputs for Capturing Fine-Scale Turbulence — *Aug 2026*
Proceedings of 2nd International Symposium on AI and Fluid Mechanics (S04: Physics-informed Models)
Amelie Lam, Héloïse Méheut, Sara El Bouch, André Ferrari

Selected Talks

AIFLUIDs 2026 – Chania, Greece — *Sep 2026*
Solver-In-The-Loop Training with Multistep Inputs for Capturing Particles in Turbulence

3iA Seminar – Sophia Antipolis, France — *Mar 2026*
CNN-Augmented Fluid Solver for Capturing Fine-Scale Turbulence

ASNUM 2025 – Grenoble, France — *Dec 2025*
Capturing Fine-Scale Turbulence in Planet-Forming Disks Using CNN-Augmented Fluid Solvers

Skills

Programming: Python, C++, Fortran, Bash

Scientific Computing: NumPy, Xarray, NetCDF, GPU Computing, HPC

Scientific Machine Learning: JAX, Flax, PyTorch, TensorFlow, CNNs, differentiable simulation

Tools: Slurm, Git, Linux